

Deep Learning Complex Computing Problem

“Can AI Save Lives? Building an Intelligent Clinical Early Warning System”

Total Marks: 10 (Implementation: 5 | Report: 5)

THE CHALLENGE

Every year, thousands of patients in hospitals deteriorate silently, their vitals shift gradually, their clinical notes grow more urgent, and yet the warning signs go unnoticed until it is too late. Hospitals are now turning to AI to build Early Warning Systems that monitor patients continuously and flag those at risk of sepsis, cardiac arrest, or ICU transfer before a crisis occurs.

You have been hired as a Machine Learning Engineer at a healthcare AI startup. Your team has been given access to a real-world clinical dataset and tasked with building a deep learning pipeline that can predict patient deterioration risk from a combination of time-series vital signs and free-text clinical notes. Your models must be accurate but more importantly, your decisions must be explainable, because in healthcare a doctor needs to understand why the model flagged a patient, not just that it did.

Dataset: Use the PhysioNet Sepsis Prediction Challenge dataset (publicly available), or the Patient Survival Prediction dataset on Kaggle. If access permits, the MIMIC-III Clinical Notes dataset is also acceptable.

You will construct three generations of models, each solving a limitation of the previous one, mirroring how the field of deep learning actually evolved.

PART A — IMPLEMENTATION

5 Marks

Generation 1: Establishing the Baseline

"Before we trust AI with patient lives, we need to know what a simple model can and cannot do."

Preprocess the dataset: handle missing vitals through imputation, normalize time-series features, and encode categorical variables. Then load and fine-tune a pre-trained DNN or shallow feedforward network (e.g., via scikit-learn's DNN Classifier or a Keras Sequential model with pre-trained weights from a similar medical dataset).

Your implementation must include:

- At least two optimizers (e.g., SGD vs. Adam), plot and compare their loss curves side by side
- Dropout and Batch Normalization as regularization strategies, report the effect of each

- Accuracy, Precision, Recall, and F1-score on the test set

Note: Given the life-critical nature of the task, pay particular attention to Recall in your notebook comments. Explain why a false negative is far more dangerous than a false positive in this context.

Generation 2: Capturing the Patient Timeline

"A patient's risk is not a snapshot, it is a story told over hours."

Restructure your data as time-series sequences (e.g., hourly vitals windows of 12–24 hours). Load and fine-tune a pre-trained LSTM or GRU model (e.g., from PyTorch Hub or a Keras model zoo entry trained on sequential medical data, or initialize with pre-trained weights and fine-tune on your dataset).

Your implementation must include:

- A fine-tuned LSTM and a GRU, report performance and training time differences
- A Bidirectional LSTM variant, evaluate whether it improves over the unidirectional version
- Training vs. validation loss curves for each variant
- A notebook comment justifying which architecture is appropriate for real-time monitoring vs. retrospective analysis and why

Generation 3: Reading the Clinical Notes

"Vitals tell you numbers. Notes tell you the story. The best systems read both."

Load a pre-trained Transformer-based model: BioBERT, ClinicalBERT, or PubMedBERT (all available via HuggingFace transformers) and fine-tune it on the free-text clinical notes in your dataset for deterioration risk classification.

You do not need to implement attention from scratch. However, your implementation must include:

- Correct tokenizer configuration and input formatting for clinical text
- Two fine-tuning strategies: frozen base with trainable head only, and full fine-tuning. Compare performance and training cost between both
- Attention weight extraction and visualization from the final attention layer (e.g., using BertViz or a manual heatmap), identify which clinical terms the model focuses on most
- A classification head appropriate for binary risk prediction
- A per-class performance breakdown: how well does the model detect true deterioration cases vs. stable patients?

Unified Model Comparison

Report all six models in the following table and plot confusion matrices for all models side by side.

Model	Accuracy	Precision	Recall	F1-Score	Training Time
DNN (Baseline)					
LSTM					
Bi-LSTM					
GRU					
ClinicalBERT (Frozen)					
ClinicalBERT (Full Fine-tune)					

PART B — REPORT

5 Marks

Your report should read like a professional engineering document addressed to two audiences simultaneously: a Chief Medical Officer who needs to trust the system, and a CTO who needs to deploy it.

Section 1: The Clinical Problem

Frame the problem for a non-technical audience and address the following:

- What makes patient deterioration prediction difficult compared to a standard classification task?
- Why is tabular vital-signs data alone insufficient, and what does unstructured clinical text add that numbers cannot capture?
- Why does Recall matter more than Accuracy in this application? Use a concrete hypothetical patient scenario to illustrate your point.

Section 2: Baseline Design Decisions

Reflect analytically on your DNN experiments and address the following:

- Explain the vanishing gradient problem and how it manifests in deep networks. How did your choice of activation function and techniques like Batch Normalization address it?
- Compare your optimizers
- Explain the distinction between a loss function and a cost function.
- Reflect on Dropout

Section 3: Modelling Patient Timelines

Reflect on your recurrent architecture experiments and address the following:

- Explain the vanishing gradient problem specifically in the context of RNNs
- Explain how LSTM gates address this problem. Then explain the GRU design tradeoff what does it give up, and what does it gain in a clinical monitoring context?

- Justify your architectural choice between unidirectional and bidirectional LSTM for real-time monitoring. Why would bidirectionality be inappropriate for a live early warning system, even if it yields higher accuracy in offline evaluation?

Section 4: The Transformer and Clinical Language

This is the analytical core of your report. Write with precision and address each of the following:

- Why pre-trained language models? Explain what transfer learning means in the context of ClinicalBERT. What knowledge does a model pre-trained on PubMed abstracts and clinical notes bring to your task that training from scratch on a small dataset cannot?
- The self-attention advantage: a doctor reading a note does not process words in order, they scan for the most relevant terms.
- What your attention heatmaps revealed: describe which clinical terms your model attended to most strongly. Do these terms align with what a clinician would consider warning signs? What does this tell you about the model's internal reasoning?
- Frozen vs. full fine-tuning: explain the tradeoff you observed empirically. When is frozen fine-tuning preferable? Consider dataset size and computational budget and when does full fine-tuning justify its cost?
- Positional encoding and why it matters for clinical notes
- Parallelisation and scale: What does this mean for a hospital system that needs to process thousands of patient notes per hour?

Section 5: Deployment Verdict

Write a concise closing argument addressing the following:

- Which model would you recommend for deployment in a real ICU and why? Weigh accuracy, explainability, latency, and the cost of errors against each other.
- What ethical concerns arise when deploying AI in a clinical setting? Consider bias, privacy and accountability.
- If the hospital wanted to expand the system to also analyze chest X-rays alongside clinical notes, how would the architecture need to evolve at a high level? A conceptual answer is sufficient, no implementation required.

DELIVERABLES

Deliverable	Requirements
notebook.ipynb	Full implementation with inline comments explaining every architectural, hyperparameter, and preprocessing decision
report.pdf	3–4 pages of analytical prose covering all five sections of Part B. Write in connected paragraphs.